

AROUND THE GAUSS POINT IN DIMENSION 2

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1. THE BERKOVICH AFFINE PLANE

Let p be a prime number. We denote by $|\cdot|$ the p -adic norm, extended from the standard norm on $\mathbb{Q}_p$ to $\mathbb{C}_p$. Set $\mathcal{O}_p := \{z \in \mathbb{C}_p : |z| \leq 1\}$ and fix once and for all a reduction map $\mathcal{O}_p \rightarrow \overline{\mathbb{F}}_p$. For any $z \in \mathcal{O}_p$, we denote by $\tilde{z}$ its image in $\overline{\mathbb{F}}_p$.

Definition 1.1. The Berkovich affine plane $\mathbb{A}^{2,an}$ over $\mathbb{C}_p$ is the analytic spectrum of $\mathbb{C}_p[S, T]$. As a set, $\mathbb{A}^{2,an}$ is the collection of all multiplicative seminorms on $\mathbb{C}_p[S, T]$ extending the norm on $\mathbb{C}_p$. The topology in $\mathbb{A}^{2,an}$ is the weakest topology such that for all $f \in \mathbb{C}_p[S, T]$, the map $\iota_f : \mathbb{A}^{2,an} \rightarrow \mathbb{R}_{\geq 0}$ given by $\iota_f(x) = |f|_x$, is continuous.

The Gauss point $\xi \in \mathbb{A}^{2,an}$ is the norm defined as

$$\left| \sum_{i,j} a_{i,j} S^i T^j \right|_{\xi} := \max\{|a_{i,j}|\}.$$

Lemma 1.2. For any $R > 1$ and $f \in \mathbb{C}_p[S, T]$, set

$$V(f; R) = \{x \in \mathbb{A}^{2,an} : R^{-1} < |f(S, T)|_x < R\}$$

A system of neighbourhoods of the Gauss point is

$$\{V(f; R) : f \in \mathbb{C}_p[S, T], |f|_{\xi} = 1, R > 1\}.$$

Proof. The topology of $\mathbb{A}^{2,an}$ is generated by sets formed by choosing an open bounded interval $I \subseteq [0, \infty)$, an element $f \in \mathbb{C}_p[S, T]$ and setting

$$U(f, I) = \{x \in \mathbb{A}^{2,an} : |f|_x \in I\}.$$

Note that for any $\lambda \in \mathbb{C}_p^*$, we have that

$$U(\lambda f, I) = U(f, |\lambda|^{-1}I).$$

Hence, we can restrict our attention to the collection

$$\{U(f, I) : |f|_{\xi} = 1, I \subset [0, \infty) \text{ bounded open interval}\}.$$

Such a set contains the Gauss point if and only if $1 \in I$. Hence, we can restrict further to intervals of the form $I = (R^{-1}, R)$ with $R > 1$, as desired. $\square$

Lemma 1.3. Let $(D_n) \subseteq \mathbb{C}_p^2$ be a sequence of effective 0-cycles such that $\deg D_n \geq 1$ for all n . Then, the following are equivalent:

- i) $\bar{\delta}_{D_n}$ weakly converges to δ_{ξ}
- ii) For every $R > 1$ and $f \in \mathbb{C}_p[S, T]$ with $|f|_{\xi} = 1$, we have that

$$\lim_{n \rightarrow \infty} \frac{\deg(D_n \cap V(f; R))}{\deg D_n} = 1.$$

Proof. Since the measures in consideration have finite support, weak convergence is equivalent to the statement that for every neighbourhood U of ξ , it holds

$$\lim_{n \rightarrow \infty} \frac{\deg(D_n \cap U)}{\deg D_n} = 1.$$

Hence, the assertion follows from Lemma 1.2. $\square$

Lemma 1.4. *Let $(D_n) \subseteq \mathcal{O}_p^2$ be a sequence of effective 0-cycles such that for any curve $C \subseteq \mathbb{A}_{\mathbb{F}_p}^2$, it holds*

$$\lim_{n \rightarrow \infty} \frac{\deg(\tilde{D}_n \cap C(\overline{\mathbb{F}_p}))}{\deg D_n} = 0.$$

Then, $\bar{\delta}_{D_n}$ weakly converges to δ_ξ .

2. HECKE CORRESPONDENCES ON THE BAD LOCUS

Let Y be the modular curve. Let ℓ be a prime number different from p and consider the Hecke correspondence T_ℓ on $Y \times Y$. In modular terms, we have that for any pair of elliptic curves (E_1, E_2) it holds

$$T_\ell(E_1, E_2) = \sum_{\substack{C_1 \leq E_1 \\ |C_1| = \ell}} \sum_{\substack{C_2 \leq E_2 \\ |C_2| = \ell}} (E_1/C_1, E_2/C_2).$$

In particular, $\deg T_\ell = (\ell + 1)^2$.

Identify $Y(\mathbb{C}_p)$ with $\mathbb{C}_p$ through the j -invariant. Then, the analytification of $Y_{\mathbb{C}_p} \times Y_{\mathbb{C}_p}$ is isomorphic to $\mathbb{A}^{2,an}$. Let Y_{bad} be set of points in $Y(\mathbb{C}_p)$ parametrizing elliptic curves with bad multiplicative reduction. Then, Hecke correspondences preserve $Y_{bad} \times Y_{bad}$.

Proposition 2.1. *Let $P \in Y_{bad} \times Y_{bad}$. Then, we have that $\bar{\delta}_{T_\ell(P)} \rightarrow \delta_\xi$ weakly as $\ell \rightarrow \infty$ among prime numbers.*

Before proving Proposition 2.1 we state an elementary fact.

Lemma 2.2. *Let m, A be positive integers and let $q \in \mathbb{C}_p$ with $|q| = 1$. Then, there exist $t \geq 1$ such that for all integers $\ell \geq 1$ with $p \nmid \ell$ it holds*

$$(2.1) \quad |\{\gamma \in \mathbb{C}_p : \gamma^{m\ell} = q \text{ and } [\mathbb{F}_p(\tilde{\gamma}) : \mathbb{F}_p] \leq A\}| \leq t.$$

*Proof.*¹ Set $n := \ell m$. All roots of $X^n = q$ lie in $\mathcal{O}_p$ because $|q| = 1$. Choose one root γ_0 . Then, the set of all roots is $\{\gamma_0 \zeta : \zeta \in \mu_n\}$, where μ_n stands for the group of n -th roots of unity in $\mathbb{C}_p$.

Write $m = p^r m_0$ with $p \nmid m_0$ and set $n' := \ell m_0$. The restriction of the reduction map $\rho : \mu_n \rightarrow \overline{\mathbb{F}_p}^\times$ is a homomorphism. Since $p \nmid \ell$, its kernel consists exactly of the p^r -th roots of unity, and its image is $\mu_{n'}$. Consequently, for a fixed $y \in \overline{\mathbb{F}_p}^\times$, the equation $\rho(\zeta) = y$ has either 0 or p^r solutions $\zeta \in \mu_n$.

Let

$$R = \{y \in \overline{\mathbb{F}_p}^\times : y^n = \tilde{q}\}, \quad S = \{y \in \overline{\mathbb{F}_p}^\times : [\mathbb{F}_p(y) : \mathbb{F}_p] \leq A\},$$

$$T = \{\gamma \in \mathbb{C}_p : \gamma^n = q \text{ and } [\mathbb{F}_p(\tilde{\gamma}) : \mathbb{F}_p] \leq A\}.$$

For any $\gamma \in T$, its reduction $\tilde{\gamma}$ lies in $S \cap R$. The set R is precisely the coset $\tilde{\gamma}_0 \mu_{n'}$ and each $y \in S \cap R$ lifts to exactly p^r distinct roots of $X^n = q$. Therefore,

$$|T| \leq p^r \cdot |S \cap R| \leq p^r |S|,$$

showing (2.1) with $t = p^r |S|$. □

Proof of Proposition 2.1. The Tate uniformization of elliptic curves with bad multiplicative reduction furnishes an identification of Y_{bad} with $\{q \in \mathbb{C}_p : |q| < 1\}$. In terms of the parameter q , we have the following description of T_ℓ . Let (q_1, q_2) be the parameters corresponding to (E_1, E_2) . Let

$$Q_\ell = \{(\eta_1, \eta_2) : \eta_1^\ell = q_1, \eta_2^\ell = q_2\}.$$

Then,

$$(2.2) \quad T_\ell(q_1, q_2) = \sum_{(\eta_1, \eta_2) \in Q_\ell} (\eta_1, \eta_2) + \sum_{\eta_1^\ell = q_1} (\eta_1, q_2^\ell) + \sum_{\eta_2^\ell = q_2} (q_1^\ell, \eta_2) + (q_1^\ell, q_2^\ell).$$

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¹Perhaps one can simplify the exposition a bit. First one notes the solutions are in $\mathcal{O}_p$. I think that it logically follows – just from the statement – that the lemma is true for a general m if and only if it is true for m of the form p^r . Then, as the map $\gamma \rightarrow \gamma^{p^r}$ is finite to 1, and $\mathbb{F}_p(\tilde{\gamma}) = \mathbb{F}_p(\tilde{\gamma}^{p^r})$, one may assume $m = p^r = 1$. Then one gives your proof, but it might be easier to follow now. RM: you are right. If we end publishing this particular piece of the problem I will do the modification you suggest (A side remark is that one usually uses q for a power of p , so it might be better to replace q by u or z , say. RM: yes, but here q is a Tate parameter)

²Is this true?? Don't you need terms like $\sum_{\eta_1 \in Q_\ell} (\eta_1, q_2^\ell)$? The sentence "We remark that..." below suggest that you simply forgot the $\sum$ sign. RM: yes, the writing of the formula was ambiguous. I corrected it.

In view of Lemma 1.3, we need to show that for every $f \in \mathbb{C}_p[S, T]$ with $|f|_\xi = 1$ and every $R > 1$, we have that

$$(2.3) \quad \lim_{\ell \rightarrow \infty} \frac{\deg(T_\ell(q_1, q_2) \cap V(f; R))}{(\ell + 1)^2} = 1.$$

We remark that in (2.2), the terms $(\eta_1, q_2^\ell) + (q_1^\ell, \eta_2)$ and (q_1^ℓ, q_2^ℓ) account for only $2\ell + 1$ points, so they have no influence in the computation of the limit in (2.3). We will focus on the terms of the form (η_1, η_2) .

Write $f(S, T) = \sum_{i,j} a_{i,j} S^i T^j$. Let³

$$I = \{(i, j) : |a_{i,j}| = 1\}, \quad \kappa := \max_{(i,j) \in I} |q_1|^i |q_2|^j, \quad C = \{(i, j) \in I : |q_1|^i |q_2|^j = \kappa\}, \quad M = \max_{(i,j) \notin I} |a_{i,j}|.$$

Assume first that C is a singleton, $C = \{(i_0, j_0)\}$. Then for every $(i, j) \notin I$, we have that $|a_{i,j} \eta_1^i \eta_2^j| < M$. For $(i, j) \in I$ with $(i, j) \neq (i_0, j_0)$, we have that

$$|a_{i,j} \eta_1^i \eta_2^j| = |q_1|^{i/\ell} |q_2|^{j/\ell} < |q_1|^{i_0/\ell} |q_2|^{j_0/\ell} = |a_{i,j} \eta_1^{i_0} \eta_2^{j_0}|.$$

Since $M < 1$ and $|a_{i,j} \eta_1^{i_0} \eta_2^{j_0}| = |q_1|^{i_0/\ell} |q_2|^{j_0/\ell}$ tends to 1 as $\ell \rightarrow \infty$, we have that for ℓ big enough it holds⁴

$$|f(\eta_1, \eta_2)| = |q_1|^{i_0/\ell} |q_2|^{j_0/\ell}$$

Hence, $|f(\eta_1, \eta_2)| \rightarrow 1$ as $\ell \rightarrow \infty$, thus establishing (2.3) in the case where C is a singleton.⁵

Now assume $|C| \geq 2$. Consider the auxiliary polynomial

$$g(S, T) = \sum_{i,j \in C} a_{i,j} S^i T^j.$$

Claim. There are positive integers α, β, k such that

- i) for any $(i, j) \in C$ it holds $k = i\alpha + j\beta$.
- ii) $\left| \frac{q_2^\alpha}{q_1^\beta} \right| = 1$.

Indeed, write $|q_i| = p^{-r_i}$ with $r_1, r_2 \in \mathbb{Q}_{>0}$. Take distinct elements $(i, j), (i', j') \in C$. Then, from $|q_1|^i |q_2|^j = |q_1|^{i'} |q_2|^{j'}$ we deduce $ir_1 + jr_2 = i'r_1 + j'r_2$. Writing $r_i = \frac{a_i}{b_i}$ with $a_i, b_i \geq 1$, we see that $\alpha = a_1 b_2, \beta = a_2 b_1$ satisfy the conditions in the claim.

Now set $h(T) := \sum_{(i,j) \in C} a_{i,j} T^{j\beta}$ and for every $(\eta_1, \eta_2) \in Q_\ell$, choose once and for all $(\xi_1, \xi_2) \in \mathbb{C}_p^2$ such that $\xi_1^\alpha = \eta_1, \xi_2^\beta = \eta_2$. Then, we see that

$$g(\eta_1, \eta_2) = g(\xi_1^\alpha, \xi_2^\beta) = \sum_{(i,j) \in C} a_{i,j} \xi_1^{i\alpha} \xi_2^{j\beta} = \xi_1^k h\left(\frac{\xi_2}{\xi_1}\right).$$

Now there are a finite number of elements $a_1, \dots, a_r \in \mathbb{C}_p$ such that $|h(z)| < 1$ if and only if $z \in \cup_{i=1}^r D^0(a_i, 1)$. So $\left| h\left(\frac{\xi_2}{\xi_1}\right) \right| < 1$ if and only if there is some $i \in \{1, \dots, r\}$ such that

$$\left| \frac{\xi_2}{\xi_1} - a_i \right| < 1.$$

Since

$$\left(\frac{\xi_2}{\xi_1} \right)^{\ell\alpha\beta} = \frac{q_2^\alpha}{q_1^\beta} =: q$$

and $|q| = 1$ due to the choice of α, β , using Lemma 2.2 we see that there is $t \geq 1$ such that for each $i = 1, 2, \dots, r$ and all $\ell \neq p$, it holds

$$\left| \left\{ (\eta_1, \eta_2) \in Q_\ell : \left| \frac{\xi_2}{\xi_1} - a_i \right| < 1 \right\} \right| \leq \ell t.$$

In conclusion, for all $\ell \neq p$ there exists a subset $R_\ell \subset Q_\ell$ such that

- i) $|R_\ell| \geq |Q_\ell| - \ell r t = \ell^2 - \ell r t$
- ii) for all $(\eta_1, \eta_2) \in R_\ell$, it holds $|g(\eta_1, \eta_2)| = |\xi_1|^k = |q_1|^{\frac{k}{\ell\alpha}}$

³Using T in the following displayed line is really bad notation RM: oh yes

⁴You write "we have that for ℓ big enough it holds.." but then write a limit... RM: I agree it's too informal, I hope now it's better

⁵Is this where you want to finish the discussion, or do you want already to conclude (2.3) at this point?? RM: this proves (2.3) in this case, I hope now it's more clear

Since

$$|f(\eta_1, \eta_2) - g(\eta_1, \eta_2)| < M < 1$$

and $|q_1|^{\frac{k}{\ell\alpha}}$ tends to 1 as $\ell \rightarrow \infty$, we conclude that for ℓ big enough

$$\frac{\deg(T_\ell(q_1, q_2) \cap V(f; R))}{(\ell + 1)^2} \geq \frac{\ell^2 - r\ell t}{(\ell + 1)^2},$$

and so (2.3) holds, as desired. □

REFERENCES

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